DATABASE MANAGEMENT SYSTEM

Project Proposal

Cat Resort Management System

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1. INTRODUCTION

The Cat Resort Management System is a **database-driven solution** that manages key operations of a cat boarding facility, including cat registration, owner management, bookings, room allocation, feeding schedules, health records, and billing.

It replaces manual recordkeeping with a **centralized database** that ensures **data consistency, accuracy, and easy retrieval**. The system supports multiple user roles like administrators, staff, and veterinarians—for real-time updates and secure access.

It also generates reports on occupancy, revenue, and health statistics, sends automated reminders, maintains secure data backups, and integrates with online booking and payment systems to enhance **efficiency and customer convenience**.

2. PROBLEMS

Current spreadsheet-based methods of recordkeeping create several issues for cat resort management:

- Data redundancy and inconsistency due to manual input.
- Difficulty in tracking room availability and booking schedules.
- Poor visibility into cats' health, vaccination, and feeding records.
- Inefficient billing and report generation.
- Risk of data loss and lack of secure access control.

A robust DBMS is required to automate data storage, retrieval, and integrity while maintaining scalability and security.

3. FUNCTIONAL REQUIREMENTS

The system must perform the following database operations and support the related modules:

I. **Cat and Owner Management**

- Store and manage cat details (Cat_ID, name, breed, age, health conditions, and vaccination status).
- Store and manage owner details (Owner_ID, name, contact, address).
- Link each cat to its owner via foreign keys.
- Allow CRUD operations for both entities.

II. **Booking and Room Management**

- Manage room details (Room_ID, type, status, price)
- Record booking details (Booking_ID, Booking_Date, Owner_ID, Room_ID, Payment_Status).
- Prevent double-booking using transaction management (ACID properties).
- Enable automated room status updates (occupied/available).

III. **Health and Feeding Management**

- Maintain health records (Health_Record_ID, Cat_ID, diagnosis, treatment, vet notes).
- Maintain feeding schedules and diet preferences.
- Support veterinarian access for medical updates.

IV. **Billing and Payment**

- Record payments (Payment_ID, Booking_ID, amount, method, status).
- Generate bills and store transaction history.

V. **Reporting and Analytics**

- Query-based reports (daily occupancy, revenue, health updates, overdue vaccinations).
- Generate summarized views for managers and admins.

4. USER TYPES

The following types of users will interact with the database:

1. **Administrator** – oversees the entire database, manages user accounts, performs backups, and controls data access.
2. **Staff Members** – handle day-to-day operations such as cat registration, bookings, feeding schedules, and payments.
3. **Veterinarian** – accesses and updates medical and vaccination records.
4. **Customer (Owner)** – limited external access (view cat status, booking history, and payments) if system includes a web interface.

5. USER ROLES

User Role	Database Permissions / Responsibilities
Admin	Full access (CREATE, READ, UPDATE, DELETE). Manages all database tables, user accounts, and backups.
Staff	Insert and update data for bookings, cats, owners, feeding, and payments. Restricted access to schema changes.
Veterinarian	Read and update health and vaccination tables only. No access to financial data.
Customer	Read-only access to their own cat's data and booking history (through controlled interface).

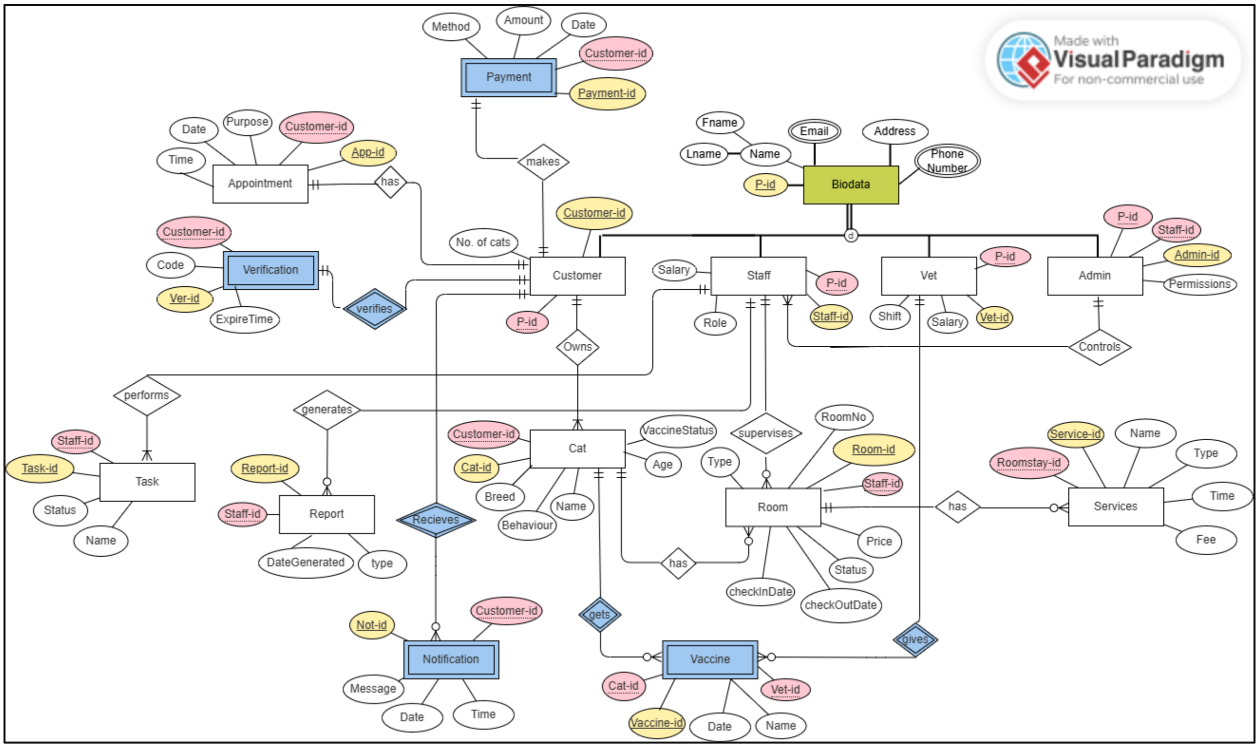
6. Entity-Relationship Diagram

1) Entities, Attributes, and Keys

Entity	Primary Key	Attributes	Foreign Keys
Customer	Customer-id	Name, No. of cats	P-id
Biodata	P-id	Fname, Lname, Email, Address, Phone Number	
Verification	Ver-id	Code, ExpireTime	Customer-id
Appointment	App-id	Date, Time, Purpose	Customer-id
Payment	Payment-id	Method, Amount, Date	Customer-id
Staff	Staff-id	Name, Role, Salary, Shift	P-id, Admin-id
Admin	Admin-id	Permissions	P-id
Vet	Vet-id	Name, Salary, Shift	P-id
Cat	Cat-id	Name, Breed, Age, Behaviour, VaccineStatus, Type	Customer-id
Room	Room-id	RoomNo, Type, Price, Status, CheckInDate, CheckOutDate	Staff-id
Services	Service-id	Name, Type, Time, Fee	Roomstay-id
Task	Task-id	Name, Status	Staff-id
Report	Report-id	Type, DateGenerated	Staff-id
Notification	Not-id	Message, Date, Time	Customer-id
Vaccine	Vaccine-id	Name, Date	Cat-id, Vet-id

2) Relationships, Cardinality, and Modality

Relationship	Entities Involved	Cardinality	Modality	Description
Owns	Customer ↔ Cat	1 : M	Mandatory on Cat side	Each customer owns many cats.
Verifies	Customer ↔ Verification	1 : 1	Optional	Verification tied to a single customer.
Makes	Customer ↔ Payment	1 : M	Mandatory	A customer can make many payments.
Has	Customer ↔ Appointment	1: 1	Mandatory	Every appointment belongs to one customer
Controls	Admin ↔ Staff	1 : M	Mandatory on Staff	Each admin controls one or more staff members.
Supervises	Staff ↔ Room	1 : M	Mandatory	Staff oversees multiple rooms.
Performs	Staff ↔ Task	1 : M	Mandatory on Task	Each staff performs many tasks.
Generates	Staff ↔ Report	1 : M	Optional	Staff generates system reports.
Receives	Customer ↔ Notification	1 : M	Optional	Customer receives system notifications.
Gets	Cat ↔ Vaccine	1 : M	Optional	Each cat can have multiple vaccines.
Gives	Vet ↔ Vaccine	1 : M	Optional	Vets administer vaccinations.
Has	Room ↔ Services	1 : M	Mandatory on Services	Services are attached to a specific stay.



7. CONCLUSION

The proposed Cat Resort Management System Database will centralize and secure all operational data for efficient management. It will minimize redundancy, enhance accuracy, and support multi-user access while ensuring data integrity and security.

This DBMS is essential for digital transformation of the cat resort's operations, enabling scalability, automation, and real-time service monitoring.